**Problem 3: Finding Complexity using Counter Method**

**Convert the following algorithm into a program and find its time complexity using counter method.**

**Factor(num) {**

```
{
  for (i = 1; i <= num;++i)
  {
   if (num % i== 0)
     {
       printf("%d ", i);
     }
   }
 }
```

**Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.**

**Input:**
**A positive Integer n**
**Output:**
**Print the value of the counter variable**

**Answer**

```
#include<stdio.h>

void Factor(int num){

   int c=0;

   int i;

  for (i = 1; i <= num;++i){

  c++;

  c++;


   if (num % i== 0)

     {

       c++;

     }

  }

  c++;
```

```
    printf("%d",c);
 }
int main(){
   int num;
   scanf("%d",&num);
   Factor(num);
   return 0;
}
```

**Feedback**

| | Input | Expected | Got | |
|---|---|---|---|---|
| | 12 | 31 | 31 | |
| | 25 | 54 | 54 | |
| | 4 | 12 | 12 | |

**Passed all tests!**

**Correct**

**Marks for this submission: 1.00/1.00.**